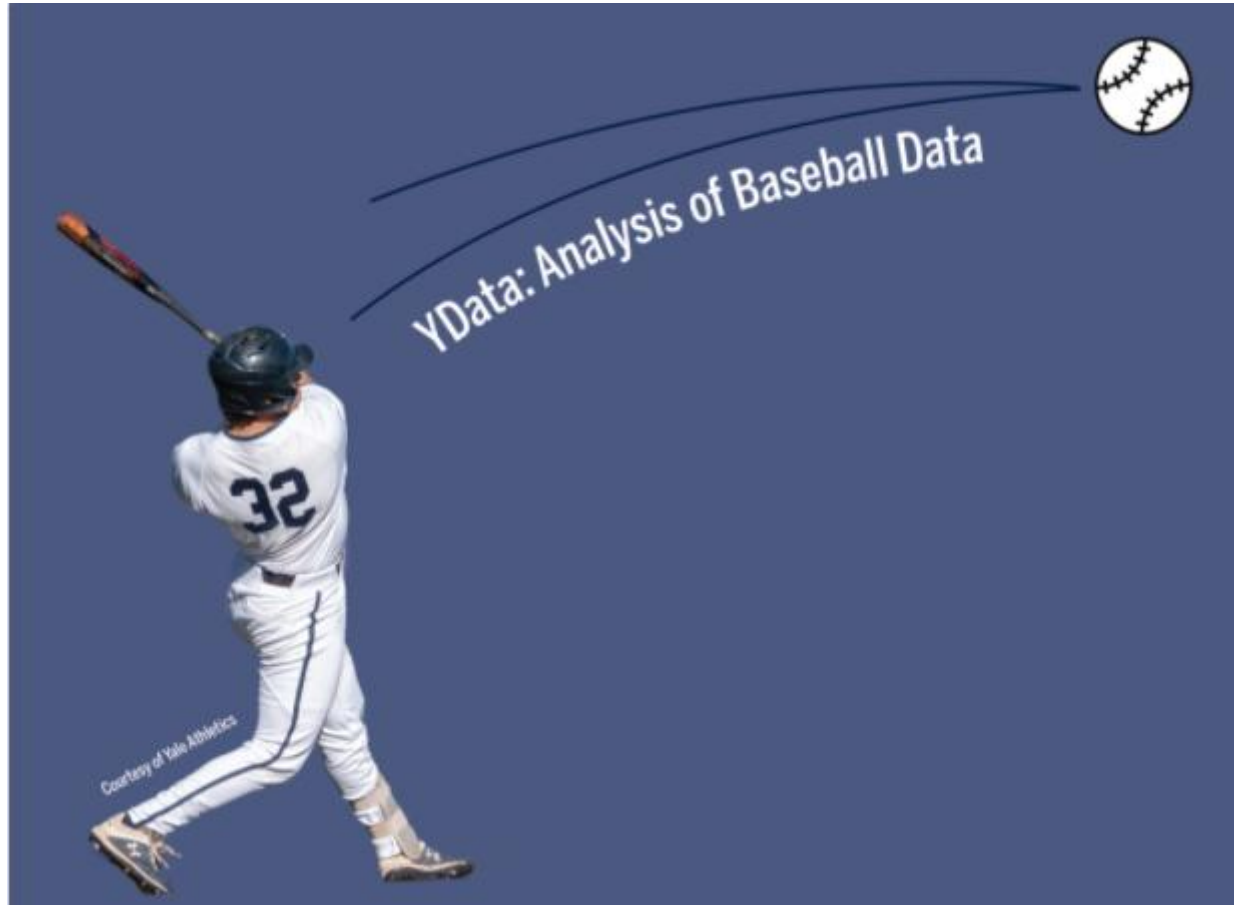


S&DS 1730



Class 1: Introductions

Overview

Course overview

- Introductions
- Syllabus and logistics

The history of statistics in baseball

Baseball statistics

If there is time:

- Quick review of pandas
- Analyzing season level statistics using pandas and the Lahman database

Office hours and contact information

Ethan Meyers (he/him)

Email: ethan.meyers@yale.edu

Office hours:

- Wednesdays and Fridays 3-4pm

Office: Kline Tower, room 1253

Teaching Assistants

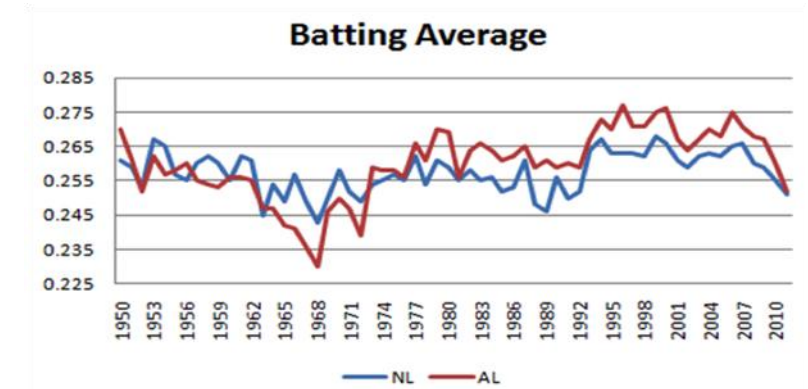
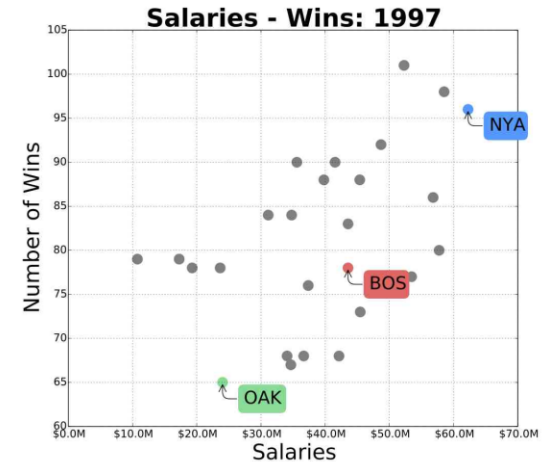
Hien Tran: hien.tran@yale.edu

Hien's office hours will be posted soon

Learning goals

1. To explore statistical/data science concepts by analyzing real and simulated baseball data
2. To gain experience using Python to analyze, visualize and wrangle data
3. To understand how Major League Baseball teams are using Data Science to improve their chances of winning

Additional goal: to have fun/socialize!



Why use baseball to study Statistics and Data Science?

High degree of randomness

- Very good players hit safely 3 out of 10 times (ave = .300)
- Bad players hit safely 2 out of 10 times (ave = .200)

Contains a rich structure that repeats, which makes it possible to isolate components and analyze them

- Discrete events makes it relatively easy to analyze:
 - Pitches -> plate appearances -> innings -> games -> seasons

Lots of data available

- Data going back to 1871

Overall: Excellent system to practice using data to answer real questions



Prerequisites

YData (S&DS 1230) or be familiar with the Python Data Science packages

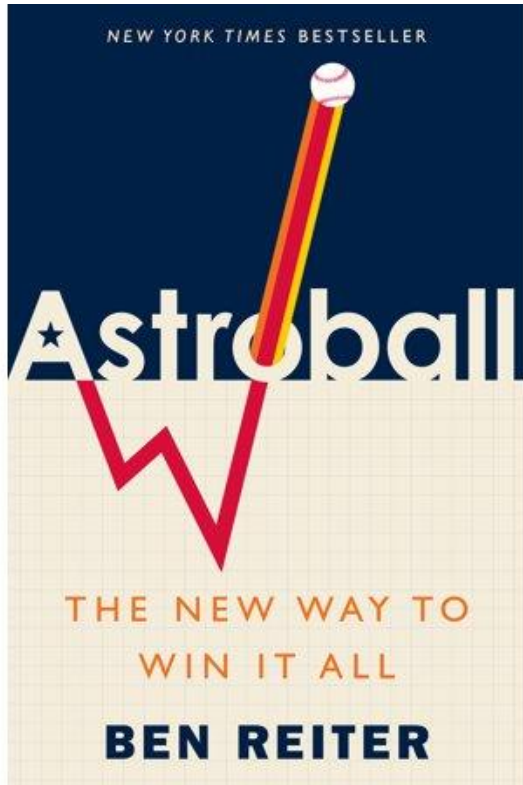
- E.g., Pandas, matplotlib, etc.

Some familiarity with baseball

- If you are not familiar at all with baseball talk to me and we will figure out a time to go over the basics

No other background is assumed

Reading: Astrobball by Reiter

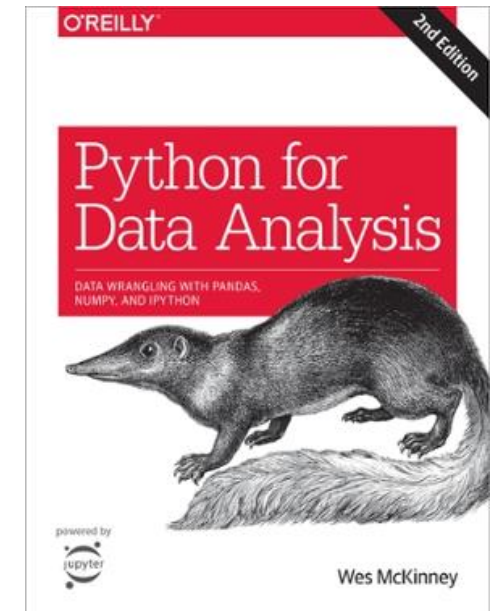
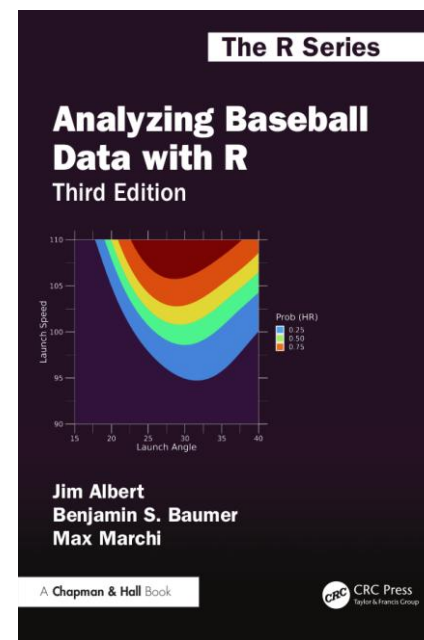
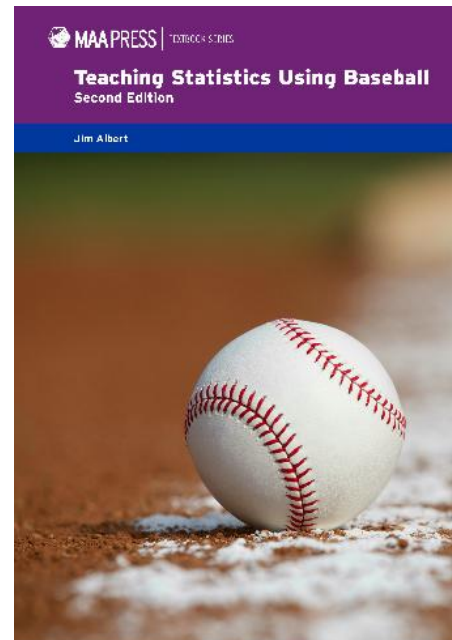
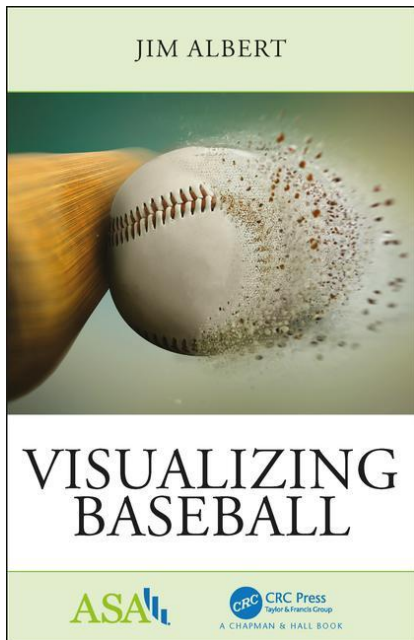


Addition reading and other resources will be posted to Canvas:

<https://yale.instructure.com/courses/115646>

Other relevant books

- Visualizing Baseball
- Teaching Statistics Using Baseball
- Analyzing Baseball Data with R
- Python for Data Analysis



Class structure

We will start with a discussion of Astroball

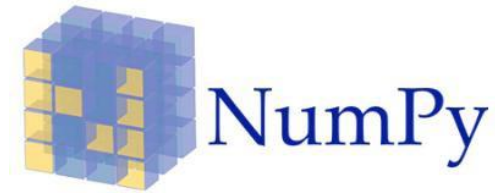
We will then cover material on a particular topic related to analyzing baseball data

The last part of the class will be an opportunity to work on a Jupyter notebook “lab”

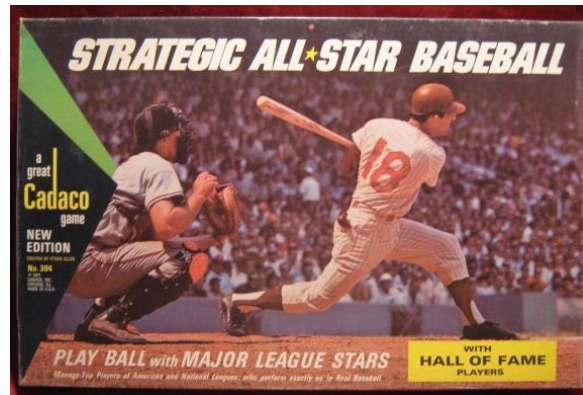
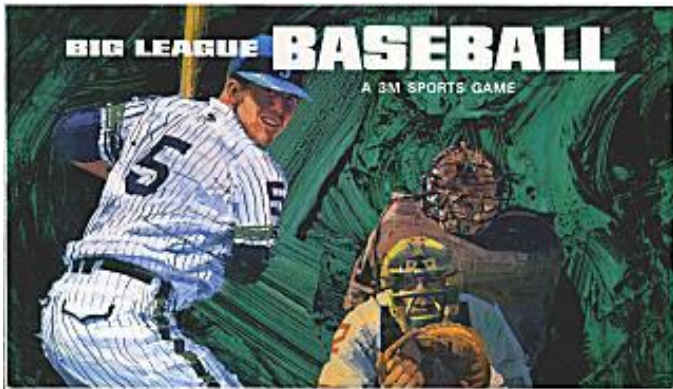
- You will finish the lab as the homework for the week

Planned topics covered

1. Practice analyzing data with Python: Descriptive statistics, data wrangling and visualization



2. Probability models and simulations using tabletop games

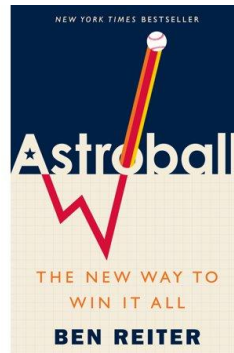


Planned topics covered

3. Inferential Statistics

- Hypothesis tests and confidence intervals

4. Linear regression classification



5. Other topics we might discuss:

- SQL
- Getting data using web APIs
- Markov chains

Types of questions we will be trying to answer

How much more valuable is a home run compared to a single?

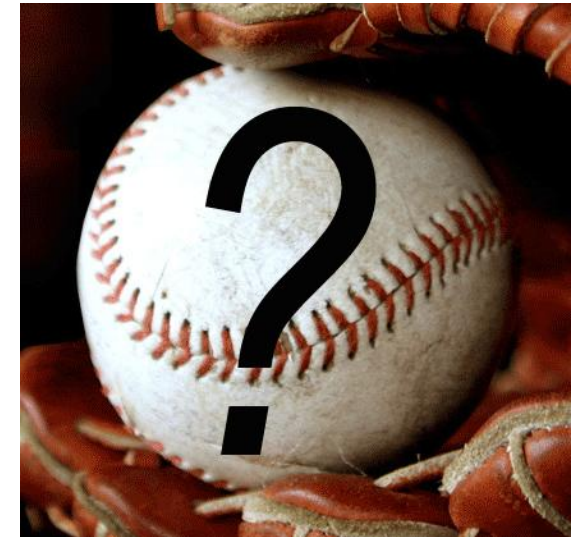
Who is the best baseball hitter of all time?

Which statistics best capture a baseball players ability?

- i.e., is on base percentage a better measure than batter average?

Are certain baseball players streaky or clutch hitters?

etc.



Assignments and grades

1. Labs/homework/problem sets (40%)
 - Exploring questions in baseball using Python
 - Weekly: 9 total, lowest score will be dropped

Lab policies:

- You may discuss questions with other but the work you turn in must be your own
- Labs will be started in class on Mondays and are due at 11pm on Sundays
- Late labs (90%) credit if turned in by 11pm on Monday

Assignments and grades

2. Final project (15%)

- This will be an opportunity to explore a question related to baseball in more depth; 8-10 page paper.

3. Exams (45%)

- Midterm: March 2nd (15% of grade)
- Final Exam: May 4th at 2pm (30% of grade)

Accommodations and Academic honesty

Accommodation: please let me know if you have accommodations for homework and/or exams

Plagiarism/cheating

- [Yale's Academic Integrity Statement](#)

You are allowed to talk with others about the homework, but the work you turn in must be your own

- Do not share answers
- Do not copy answers off the Internet
- Do not look at past year's homework

ChatGPT and other LLMs

You can use LLMs as a reference

- E.g., "What is the function to do x?"
 - i.e., ok to use it like Google/Stack Overflow

Do not use it to answer lab/homework questions

- i.e., Do not type a lab/homework question into ChatGPT
- Do not have it complete code you started writing (or cut and paste any code from an LLM)

If it appears your homework answers were generated by ChatGPT or another LLM (Claude, Gemini, etc.) you will be referred [Yale Executive Committee](#)

- Also, if you cheat on the labs/homework you will do very poorly on the exams

Class survey

In order to get to know you and to adjust the class to everyone's interests, please fill out the class survey on canvas

- Under the Quizzes link on the left

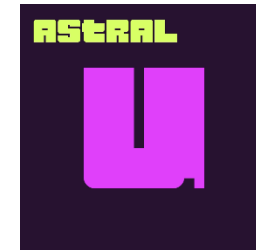
Any questions about the class logistics???

Running Jupyter Notebooks

In order to do the homework, you will need to be able to run Jupyter Notebooks

The strongly preferred way to do this is to use the YCRC Jupyter server

Before we get started, let's test the [YCRC Jupyter notebook server](#)



The history of baseball statistics



Early statistics

Henry Chadwick (1824-1908) created the first box score in the 1859 issue of Clipper

- First to use K for strike outs, helped invent batting average and earned run average
- Did not record walks because he did not feel they reflected a batter's skill



BOSTON.						ATHLETIC.										
G. Wright, s.s.	6	4	4	1	5	2	Force, s.s.	5	1	2	1	3	2			
Leonard, 2b.	6	3	3	4	4	3	Kagler, c. f.	5	3	3	0	0	0			
O'Rourke, 1b	6	2	3	9	0	1	Fisher, r. f.	5	0	1	2	0	0			
Murnan, l. i.	6	1	0	3	1	0	Meyerle, 3db	5	1	2	2	3	3			
Schafer, 2d b	6	3	3	3	1	2	Sutton, 1st b.	5	1	2	10	0	0			
McGinley, c. f	6	0	0	0	0	1	Coons, c.	5	1	0	1	1	3			
Manning, r. f.	6	0	2	2	0	0	Hall, l. f.	5	1	3	5	0	0			
Morrill, c.	6	2	2	4	1	2	Fowler, 2d b.	6	1	2	6	7	5			
Josephs, p. . .	5	4	4	1	1	2	Knight, p. . .	5	2	2	0	1	2			
Totals....53						19	21	27	13	13	Totals....46					
Boston.....						0	1	3	3	4	1					
Athletic.....						1	0	0	0	3	3					

Runs earned—Boston, 4; Athletic, 5. Home-run—Hall, 1.
Total bases on hits—Boston, 22; Athletic, 20. First base by
errors—Boston, 8; Athletic, 5. Umpire, George White of
Lowell, Mass. Time 2h. 47m.

Early statistics



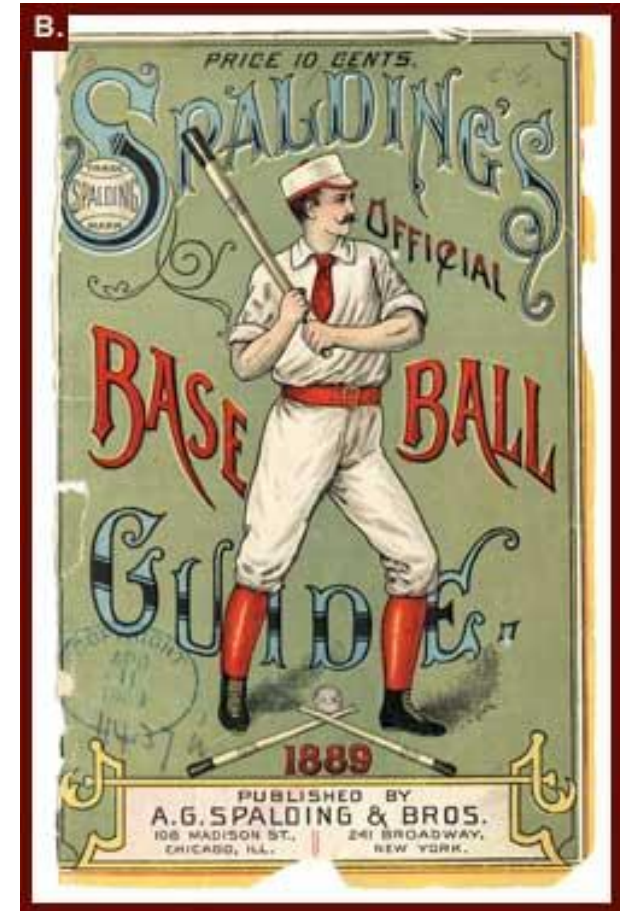
Classic statistics

Most prominent hitting statistics:

- Batting average, RBIs, and home runs

Most prominent pitching statistics:

- Wins, earned run average, strike outs



Sabermetrics

Around 1970 Bill James, and others began to question how useful traditional measures of performance

- i.e., are batting average, pitcher wins, etc. the best ways to tell how good a player is?

Sabermetrics definitions:

1. The empirical or mathematical/**Statistical** study of baseball
2. "the search for objective knowledge about baseball"

- [Bill James](#)

Name comes from 'Society for American Baseball Research' (SABR), a group started in 1971

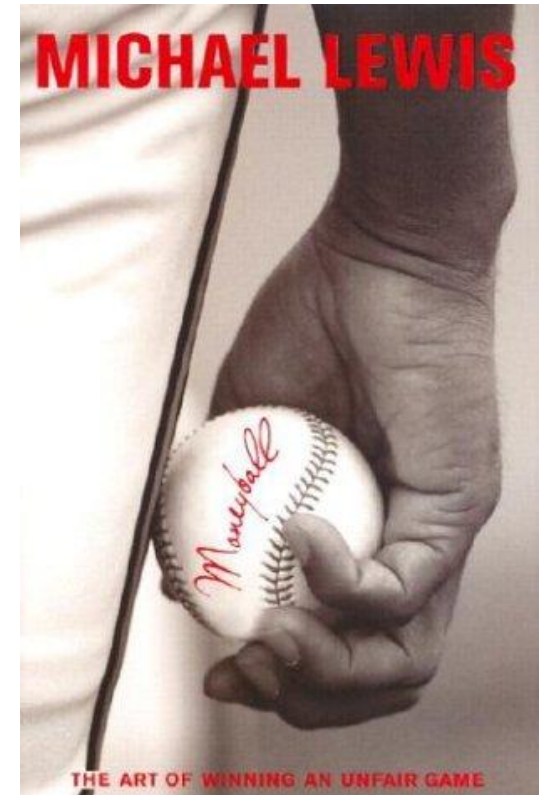
- Pre-computers, so they **had to compile all information from box scores by hand since there was no encyclopedia that had pitch-by-pitch data**

Sabermetrics first widely introduced to the public in 1982 with the publication of *Bill James Baseball Abstract*

Moneyball

Story about how the Billy Bean, the general manager of the A's, was able to put together a top ranked team in 2002 on a tight budget by finding undervalued players using advanced statistics

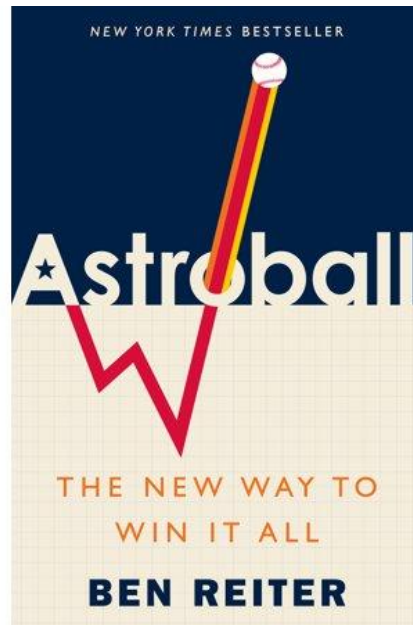
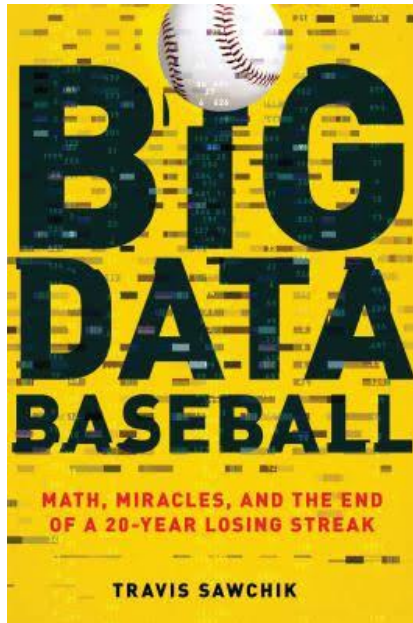
Some of the claims in the book might be exaggerated but it had a big impact on the expansion of major league clubs doing advanced data analyses



Sabermetrics continues to advance

Several books have been written about more recent sabermetric advances

- 2013 Pittsburgh Pirates
- 2017 Astros



A few prominent Sabermetric publications/websites

[Society for American Baseball Research](#)

[Bill James Online](#)

[Baseball Analysts](#)

[Baseball Prospectus](#)

[Beyond the Box Score](#)

[Fan Graphs](#)

[The Hardball Times](#)

[Tango Tiger](#)



Baseball data sets

Lahman Database: Season-by-season data

Retrosheet Game-by-Game data

Retrosheet Play-by-Play data

PITCHf/x: Pitch-by-Pitch location, pitch type data (2006)

Statcast: high-accuracy tracking of player movements (2015)



Statcast: MLBAM @ 2014 MIT Sloan Analytics Conference

Analyzing a catch by Jason Heyward



Using cognitive neuroscience to improve player performance

Inside the Mind of a Hitter

A look at how quickly a hitter must assess and react to a 90-mph fastball.

 'In the blink of an eye'
A voluntary blink—such as one caused by the flash of a light—takes about 150 milliseconds. A 90-mile-per-hour fastball will cross the plate in under three blinks

The wind-up

Hitter attempts to get first insight on the type of pitch by watching the pitcher's hand as he releases the ball.

75-100 milliseconds

After the pitcher releases his pitch, the ball travels about 9 feet before the batter is able to process the entire image of the pitcher's wind-up and release.

175

Hitter assesses the type of pitch, extracting meaningful information about velocity, spin and trajectory.

225

Decision time—to swing or not to swing? It takes around 150 milliseconds from the start of a swing to the time it makes contact with the ball, but the decision must be made around 25 milliseconds earlier, to allow time for the brain signals to reach the various muscles involved.

350

At this point only exceptional hitters can make small adjustments. The bat is traveling at about three-fourths of its final velocity.

400

The ball crosses the front of home plate.

Experience helps the hitter 'pattern' the type of throw the pitcher is known for to speed up his thought process. Is the pitcher a known fireballer? If yes, the fastball will be at the top of the list when the batter gets his first visual clue from the ball.

Having decided the character of pitch, the hitter selects a **swing pattern** that was established through countless hours of practice and experience. For this pitch he may choose 'upward swing to send the ball over the fence.'

If the swing is as little as seven milliseconds late, his squarely hit ball will go foul.

Source: 'The Physics of Baseball,' by Robert K. Adair; 2002 edition

Mike Sudal/The Wall Street Journal

<https://www.wsj.com/articles/baseballs-science-experiment-1411135882>

Common baseball statistics

Common baseball statistics

Let's look at some baseball cards...





HT: 6'0" WT: 200 BATS: LEFT THROWS: RIGHT DRAFT: ROYALS #2-JUNE, 1971
 ACC: VIA DRAFT BORN: 5-15-53, GLEN DALE, WEST VIRGINIA HOME: RANCHO MIRAGE, CALIF.

GEORGE BRETT • 1B

COMPLETE MAJOR LEAGUE BATTING RECORD (LEAGUE LEADER IN ITALICS, TIE ♦)

YR	CLUB	G	AB	R	H	2B	3B	HR	RBI	SB	SLG	BB	SO	AVG
73	ROYALS	133	40	2	5	2	0	0	0	0	.175	0	5	.125
74	ROYALS	133	457	49	129	21	5	2	47	8	.363	21	38	.282
75	ROYALS	159	<i>634</i>	84	<i>195</i>	35	<i>13</i> ♦	11	89	13	.456	46	49	.308
76	ROYALS	159	<i>645</i>	94	<i>215</i>	34	<i>14</i>	7	67	21	.462	49	36	<i>.339</i>
77	ROYALS	139	564	105	176	32	13	22	88	14	.532	55	24	.312
78	ROYALS	128	510	79	150	<i>45</i>	8	9	62	23	.467	39	35	.294
79	ROYALS	154	645	119	<i>212</i>	42	<i>20</i>	23	107	17	.563	51	36	.329
80	ROYALS	117	449	87	175	33	9	24	118	15	<i>.664</i>	58	22	<i>.390</i>
81	ROYALS	89	347	42	109	27	7	6	43	14	.484	27	23	.314
82	ROYALS	144	552	101	166	32	9	21	82	6	.505	71	51	.310
83	ROYALS	123	464	90	144	38	2	25	93	0	<i>.563</i>	57	38	.310
84	ROYALS	104	377	42	107	21	3	13	69	0	<i>.459</i>	38	37	.284
85	ROYALS	155	550	108	184	38	5	30	112	9	<i>.565</i>	103	49	.335
86	ROYALS	124	441	70	128	28	4	16	73	1	.481	80	45	.290
87	ROYALS	115	427	71	124	18	2	22	78	6	.496	72	47	.290
88	ROYALS	157	589	90	180	42	3	24	103	14	.509	82	51	.306
89	ROYALS	124	457	67	129	26	3	12	80	14	.431	59	47	.282
90	ROYALS	142	544	82	179	<i>45</i> ♦	7	14	87	9	.515	56	63	<i>.329</i>
MAJ. LEA. TOTALS		2279	8692	1382	2707	559	127	281	1379	184	.502	964	696	.311

©MLB & MLBPA 1991 D.R. ©1991 THE TOPPS COMPANY, INC.

sstatistics and structured data

statistics: a numerical summary of data
(technically a summary of a data sample)

Statistics: is the mathematics of collecting, organizing and interpreting data

Describing and summarizing data

statistics that are used to summarize a data set (sample of data) are called **descriptive statistics**

Examples:

- Maximum value in the data set
- Minimum value in the data set
- Mean value of the data set

Common baseball descriptive **s**tatistics

G = games

- Number of games a player participated in (out of 162 games in a season)

AB = at bats

- Number of times a batter was hitting and either got a hit or an out
 - Does not include walks or reaching base on an error

R = runs

- Number of runs the player scored

H = hits

- Number of times a player hit the ball on got on base or hit a home run
- Sum of 1B, 2B, 3B, HR

Common baseball descriptive **s**tatistics

BB = walks (base on balls)

- Number of times a player got on base do to the pitcher throwing 4 balls

RBI = Runs batted in

- How many runs scored as a result of a player getting a hit

SB = stolen bases

- Number of times a runner advanced by 'stealing a base'

Common derived baseball statistics

AVG= batting average

- $AVG = \text{Hits}/(\text{At bats}) = H/AB = (1B + 2B + 3B + HR)/AB$

SLG = slugging percentage

- $SLG = (1 \cdot 1B + 2 \cdot 2B + 3 \cdot 3B + 4 \cdot 4B) / AB$

Lab 0

Let's start Python with some basic warmup exercises

- you will not turn in this lab

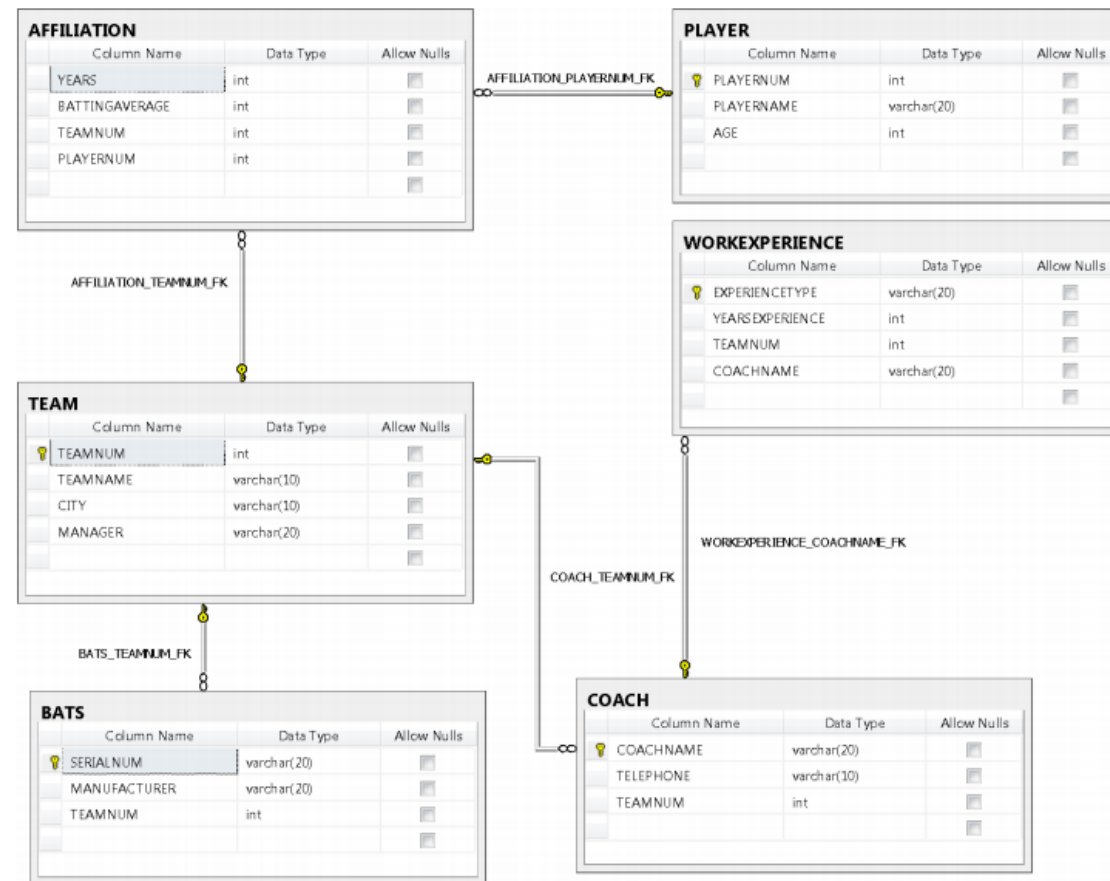
Log into the YCRC Jupyter server and create a new Jupyter notebook

Then run the following commands to get lab 0:

```
import YData_baseball as ybaseball  
ybaseball.download_lab(0)
```

Ask questions as they come up!

Lahman Database: Season level statistics



Lahman Database: Season level statistics

Lahman database contains season level information about Major League baseball players from 1871 to 2024

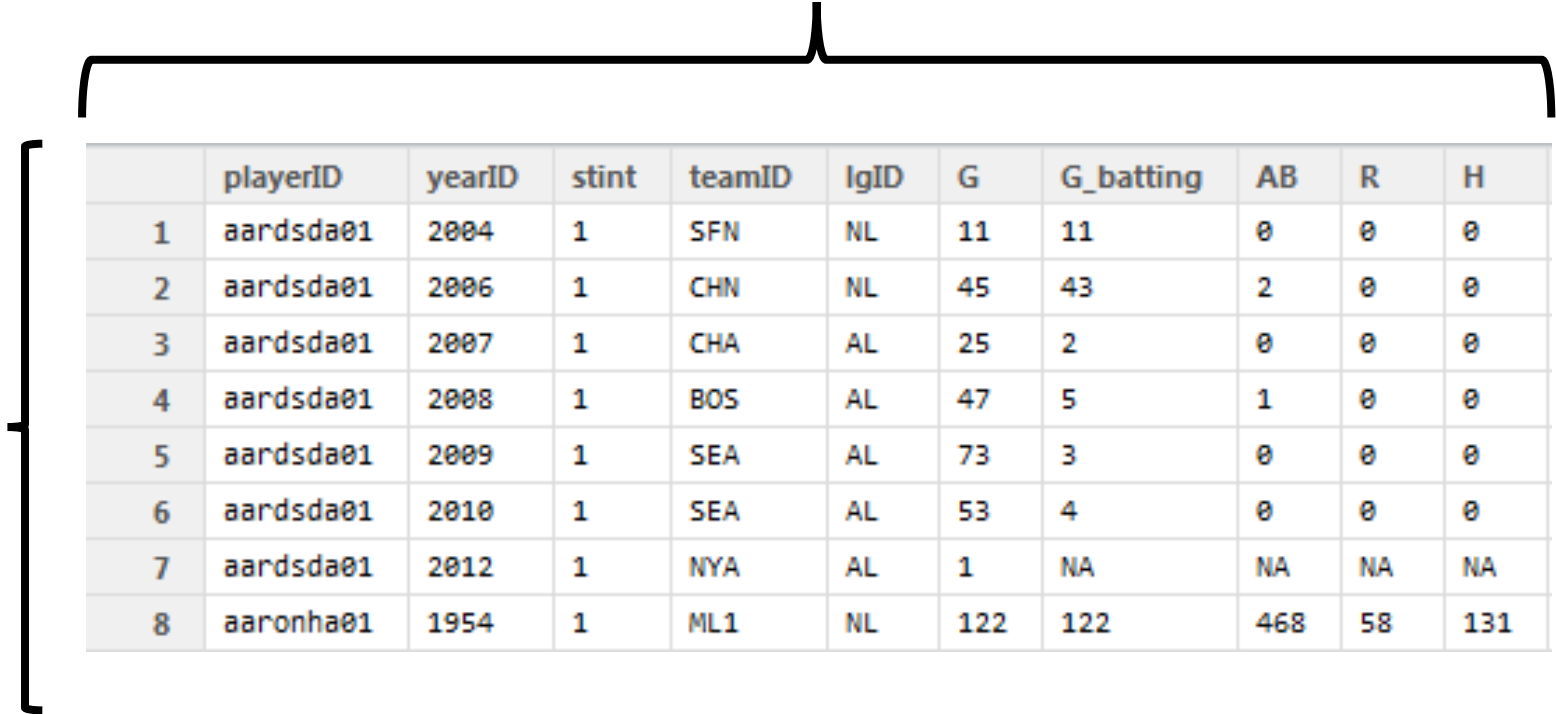
We can use this data to extract the data on your baseball card!

But first let's talk about some general terms for structured data

Structured data: Batting dataset

Variables

Cases

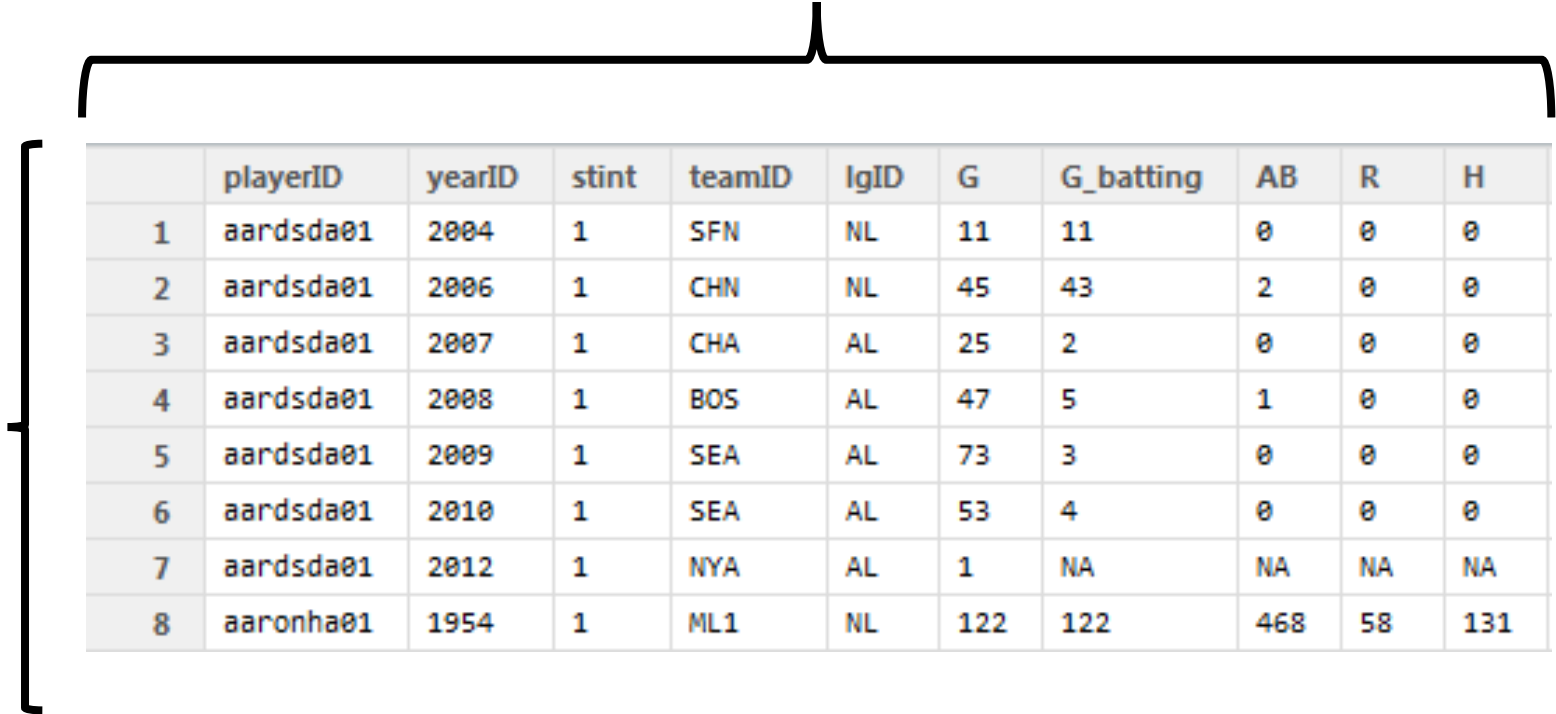


	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

Structured data

Variables

Cases



	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

Categorical and Quantitative Variables

Categorical Variable

Quantitative Variable

Cases

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

Explanatory and Response Variables

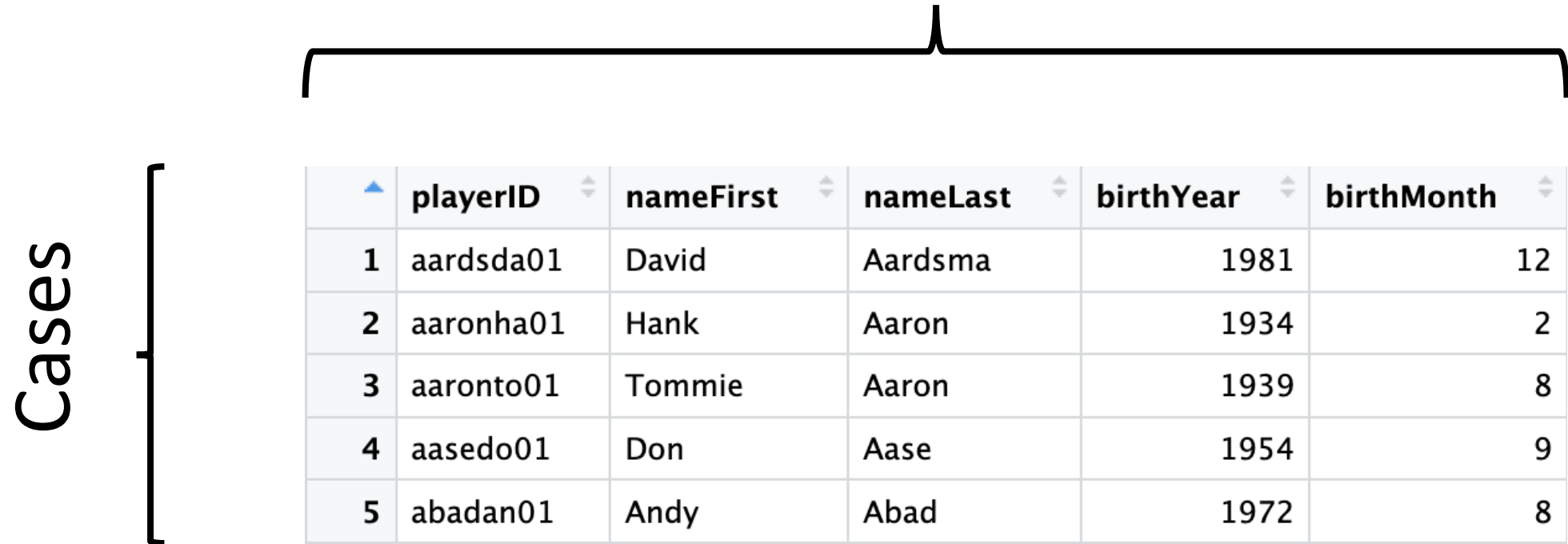
Sometimes we use one variable (**the explanatory variable**) to understand/predict another variable (**the response variable**)

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

People Dataset: Information on players

Variables

Cases



	playerID	nameFirst	nameLast	birthYear	birthMonth
1	aardsda01	David	Aardsma	1981	12
2	aaronha01	Hank	Aaron	1934	2
3	aaronto01	Tommie	Aaron	1939	8
4	aasedo01	Don	Aase	1954	9
5	abadan01	Andy	Abad	1972	8

Loading the Lahman data into Python

```
import pandas as pd
```

```
batting = pd.read_csv('Batting.csv')
```

```
players = pd.read_csv("People.csv")
```

Connecting DataFrames

The DataFrames in the Lahman database can be connected through different ID columns

- For example, we can relate the Batting and People DataFrames using the playerID column

Batting DataFrame

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

People DataFrame

	playerID	nameFirst	nameLast	birthYear	birthMonth
1	aardsda01	David	Aardsma	1981	12
2	aaronha01	Hank	Aaron	1934	2
3	aaronto01	Tommie	Aaron	1939	8
4	aasedo01	Don	Aase	1954	9
5	abadan01	Andy	Abad	1972	8

Quick review: pandas DataFrames

Quick review: pandas DataFrames

Pandas DataFrames hold Table data

	playerID	yearID	stint	teamID	lgID	G
0	aardsda01	2004	1	SFN	NL	11
1	aardsda01	2006	1	CHN	NL	45
2	aardsda01	2007	1	CHA	AL	25
3	aardsda01	2008	1	BOS	AL	47

Selecting columns:

- `my_df[["col1", "col2"]]` # getting multiple columns using a list

Extracting rows:

- `my_df.iloc[0]` # getting a row by number
- `my_df.loc["index_name"]` # getting a row by Index value
- `my_df[my_df["col_name"] == 7]` # getting rows using a Boolean mask
- `my_df.query("col_name == 7")` # getting rows using the query function

Quick review: pandas DataFrames

Sorting rows of a DataFrame

```
my_df.sort_values("col_name", ascending = False)  # sort from largest to smallest
```

Adding a new:

```
my_df["new_col"] = values_array
```

Renaming a column:

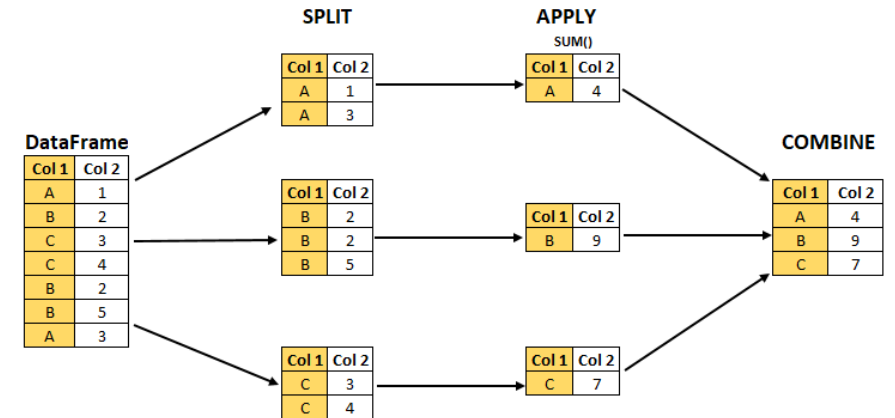
```
rename_dictionary = {"old_col_name": "new_col_name"}  
my_df.rename(columns = rename_dictionary )
```

Quick review: Creating aggregate statistics by group

We can get statistics separately by group:

```
dow.groupby("Year").agg("max")
```

```
my_df.groupby("group_col_name").agg(  
    new_col1 = ('col_name', 'statistic_name1'),  
    new_col2 = ('col_name', 'statistic_name2'),  
    new_col3 = ('col_name', 'statistic_name3')  
)
```

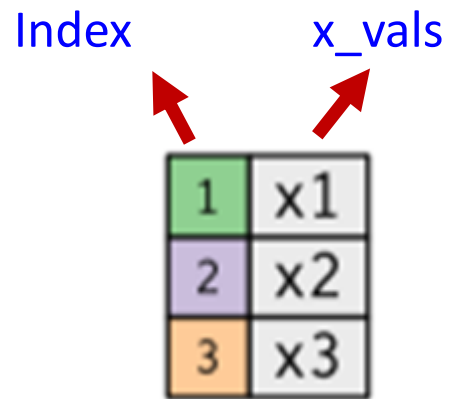


```
nba_salaries.groupby("TEAM").agg(  
    max_salary = ("SALARY", "max"),  
    min_salary = ("SALARY", "min"),  
    first_player = ("PLAYER", "min")  
)
```

Review: Joining

Suppose we have two DataFrames (or Series) called **x_df** and **y_df**

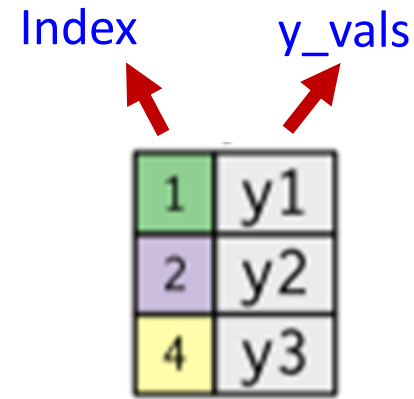
- **x_df** have one column called **x_vals**
- **y_df** has one column called **y_vals**



A diagram of a DataFrame with two columns. The first column is labeled 'Index' with a red arrow pointing to it. The second column is labeled 'x_vals' with a red arrow pointing to it. The DataFrame contains three rows: the first row has index 1 and value x1; the second row has index 2 and value x2; the third row has index 3 and value x3.

Index	x_vals
1	x1
2	x2
3	x3

DataFrame: x_df



A diagram of a DataFrame with two columns. The first column is labeled 'Index' with a red arrow pointing to it. The second column is labeled 'y_vals' with a red arrow pointing to it. The DataFrame contains three rows: the first row has index 1 and value y1; the second row has index 2 and value y2; the third row has index 4 and value y3.

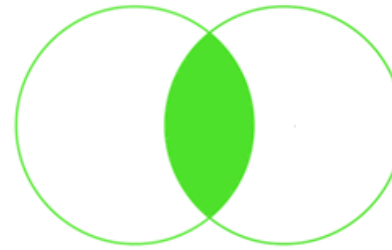
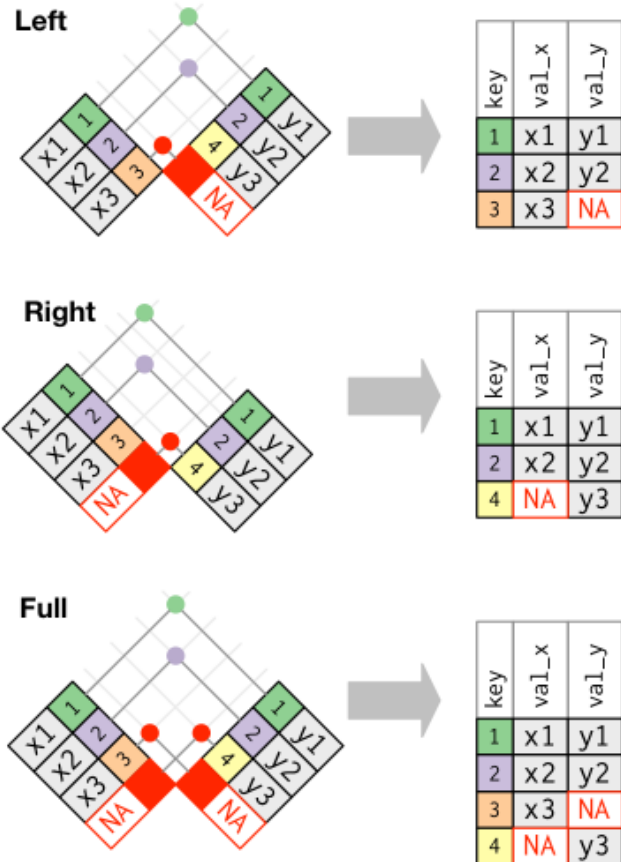
Index	y_vals
1	y1
2	y2
4	y3

DataFrame: y_df

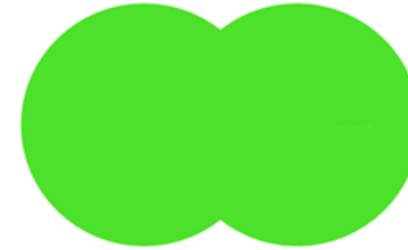
We can join these two DataFrames into a single DataFrame by aligning rows with the same Index value using the general syntax: **x_df.join(y_df)**

- i.e., the new joined data frame will have two columns: **x_vals**, and **y_vals**

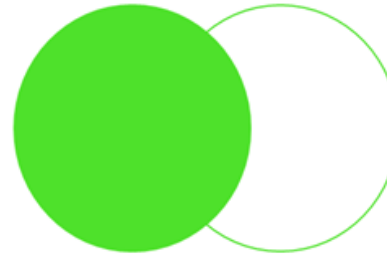
Review: Joining



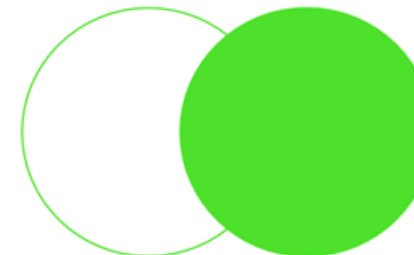
INNER JOIN



FULL OUTER JOIN



LEFT JOIN



RIGHT JOIN

`x_df.join(y_df, how = "left")` # based on Index

`x_df.merge(y_df, how = "inner", left_on = "x_col", right_on = "y_col")` # based on columns

Let's do some analyzes on the Lahman data using pandas!

```
import pandas as pd
```

```
batting = pd.read_csv('Batting.csv')
```

```
players = pd.read_csv("People.csv")
```

For next class

1. Read the preface and prologue to Astrobball
 - I will post scanned copies to the class Canvas site

2. Fill out the class background survey

3. If you are not familiar with baseball, try to learn the basics of the game
 - [Here is a short video](#) explaining the rules
 - [Here is a longer video](#)

If you're not familiar with the data science packages in Python, try to get up to speed as quickly as possible

- [This tutorial could be useful](#)

